

Enhancing Resilience through AI-Driven Virtual Reality

Talya Maness*

School of Psychology, Reichman University, Herzliya, Israel

Anat Brunstein-Klomek †

School of Psychology, Reichman University, Herzliya, Israel

Doron Friedman ‡

School of Communications,
Reichman University,
Herzliya, Israel

ABSTRACT

We introduce AI-enhanced virtual humans to address the shortage of human professionals in mental health. Specifically, we allow for automatically learning and practicing resilience skills by assisting a distressed virtual human, utilizing the established Resilience Plan Intervention (RPI). One hundred participants underwent a brief training session in the RPI and then applied their newly acquired resilience strategies in simulated interactions with a distressed individual, conducted through either VR or traditional role-playing (non-VR). The results revealed significant enhancements in reported resilience for both groups, accompanied by significant increases in reported levels of cognitive flexibility, emotion regulation, and help-seeking capabilities.

Index Terms: Virtual human, virtual reality, virtual patient, mental health, intent recognition, automated dialogue, resilience, cognitive flexibility, emotion regulation, help seeking.

1 INTRODUCTION

Resilience reflects qualities that enable thriving amidst adversity [9]. By merging AI-driven dialogue agents with VR, we aim to broaden access to personalized and immersive mental health interventions. Our innovative approach treats resilience as a muscle, enhanced through skills training based on Interpersonal Psychotherapy (IPT) and Cognitive-Behavioral Therapy (CBT). Building on prior studies that demonstrate the effectiveness of virtual training compared to traditional role-play [11, 7], this study advances the field by employing a fully automated virtual human for resilience training. We hypothesize that this system can match the efficacy of role-play methods and explore potential benefits for participants providing the training.

2 METHOD

The study included 100 **participants**: aged 18-67 ($M = 24.83$, $SD = 7.98$, 64 females), 55 were assigned to the VR group and 45 to the Role-Play group. The revised study protocol was approved by the ethical committee.

Procedure: Participants were asked to complete a set of online questionnaires before arriving at the lab for the study. The study began with the participants watching a 15-minute tutorial video explaining the theoretical basis of the resilience plan. Immediately after, they were asked to implement what they had learned, either in a role-play simulation with a person in distress, or, in the VR condition, participants put on the VR device and met “Kevin”, a 31-year-old American (Fig. 1). At the end of the intervention (Time 1) participants completed another set of questionnaires.

*e-mail: maness.talya@post.runi.ac.il

†e-mail: bkanat@runi.ac.il

‡e-mail: doronf@runi.ac.il

Virtual Human Implementation: Kevin was designed and implemented as a structured dialogue agent —while the most advanced systems can be used for natural language understanding, we restrict language generation. This allows for closely following a pre-defined transcript, at the expense of a rich open-ended dialogue. For language understanding we used wit.ai [6], a cloud service focused on intent recognition.



Figure 1: A snapshot showing the virtual distressed agent ‘Kevin’: the virtual meeting took place at the patient’s virtual home.

The scene was composed of five stages implemented as a state machine: i) introduction (getting to know Kevin), ii) inquire Kevin about his distress signals, iii) identify ways for Kevin to help himself (e.g., hobbies), iv) identify at least one concrete person that Kevin can approach to for help, and v) identify the professional bodies to approach when all else fails. Each of these stages included approximately a dozen intents trained with specific responses from Kevin; overall the system was used with 70 intents (such as “How is your daily routine?” or “Is there anyone you usually contact for support?”). Additionally, there were some automatic fillers such as “I did not understand this, can you please repeat?”.

Rigorous user testing and refinement was conducted in several phases, in order to cover most expected script paths. This was first achieved by low fidelity prototyping – role playing with an experimenter, and later followed by additional training and script adjustments within the wit.ai software. Additionally, we have developed a smartphone application for training the dialogue agent, and our logs indicate that 21 individuals (lab members) contributed to the training process, over 105 sessions.

For non-verbal behavior, we used an early version of our lab platform [8], which allows for semi-automated generation of VR scenarios with multiple virtual humans. In this case the agent had only two states: speaking and listening, and several animations were selected for each state accordingly, and the animations were intended to be ‘sad’ or similar. The gaze was selected to be 70% at the participant and 30% down (to indicate emotional distress). The overall result is a relatively simple virtual human but one that is expected to result in behavioral realism.

We ran the study on a Meta Quest 2 device connected to a laptop

running a 3070 Nvidia GPU card. The software is implemented in Unity with external cloud-based services.

The following questionnaires were administered: a demographic questionnaire, the Patient Health Questionnaire (PHQ-9; [5]), the Generalized Anxiety Disorder (GAD-7; [10]), the Connor-Davidson Resilience Scale (CD-RISC; [1]), the Cognitive Flexibility Inventory (CFI; [2]), the Difficulties in Emotion Regulation Scale (DERS; [4]), and Attitudes Toward Seeking Professional Psychological Help (ATSPPH; [3]).

3 RESULTS

A repeated measures Multivariate Analysis of Variance (MANOVA) examined the main effect of Time, Condition (VR/Role-play), and their interactions, with condition as a between-subject factor and time as a within-subject factor. As hypothesized, the analysis showed a significant multivariate time effect when predicting an increase in the mental resilience index (Fig. 2a) (Wilks' $\Lambda = 0.426$, $F(1,77) = 101.13$, $p < .001$, partial $\eta^2 = 0.51$) and cognitive flexibility (Fig. 2b) (Wilks' $\Lambda = 0.104$, $F(1,77) = 657.88$, $p < .001$, partial $\eta^2 = 0.892$). There was no significant Time $\times$ Condition interaction, neither for general mental resilience nor for cognitive flexibility (Wilks' $\Lambda = 0.88$, $F(1,77) = 9.18$, $p = .307$, partial $\eta^2 = 0.014$, respectively).

Following the MANOVA we conducted exploratory paired samples t-tests on each sub-skill per condition. This revealed a statistically significant change in emotional regulation ability from pre- to post-intervention in both groups; in the Role-Play group, $t(44) = 6.443$, $p < .001$ and in the VR group, $t(41) = 4.601$, $p < .001$. As expected, emotional dysregulation's mean pre-intervention was significantly higher ($M=32.46$, $SD=9.13$) compared to pre-intervention mean ($M= 29.15$, $SD=9.11$). Comparable outcomes were measured in the VR group, as emotional dysregulation's mean pre-intervention was higher ($M= 35.309$, $SD=10.15$) than the post-intervention mean ($M= 30.28$, $SD=10.04$).

As the MANOVA indicated that both conditions are similar and effective interventions, we have also tested for the sub-skills when the two conditions are combined. A paired-samples t-test revealed a statistically significant improvement in attitudes toward seeking professional psychological help following the intervention, with both groups (conditions) combined. The mean score at Time 0 ($M = 21.98$, $SD = 3.07$), compared to the mean score at Time 1 ($M = 28.95$, $SD = 2.97$). The analysis indicated a significant increase in positive attitudes toward seeking professional help, $p < .05$. There were no statistical differences in the other psychological measurements before and after intervention, in both groups separately and combined.

4 DISCUSSION

We observed significant enhancements in overall mental resilience, specifically in cognitive flexibility, emotion regulation, and the intent to seek help, across both conditions VR and role-play. The next generation of AI-enhanced VR presents exciting possibilities. Advanced virtual humans endowed with social intelligence into VR environments could support rich, complex social interactions tailored to the specific needs of individual users or distinct target groups. This progression will not only refine the adaptability and impact of VR interventions but may also revolutionize mental health practices by enabling more nuanced and sophisticated therapeutic interactions within virtual settings. Future research should explore the tension between structured and constrained dialogue vs a less constrained dialogue now enabled by LLMs.

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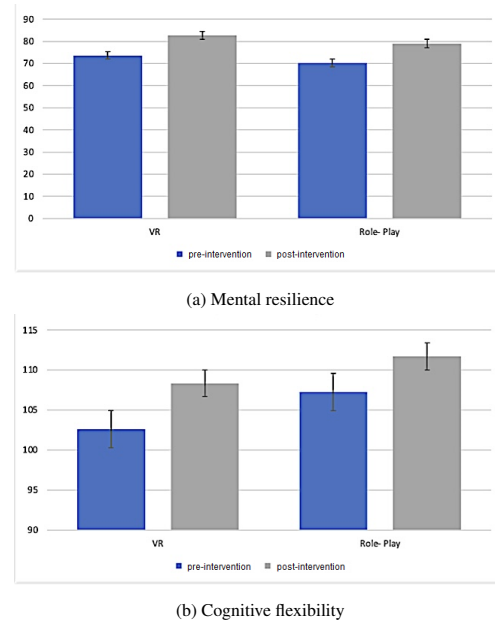


Figure 2: Bar plots displaying pre- vs post-intervention results in both conditions; error bars indicate ± 2 SE. (a) Mental resilience, (b) Cognitive flexibility.

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